

Question Paper

Exam Date & Time: 24-Nov-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FIRST SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

SUBJECT: BIO 1071/BIO_1071 BIOLOGY FOR ENGINEERS

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Justify how the chemical bonding characteristics of silicon influence its ability to support life in extraterrestrial alien environments. You need to write the answer in the following format: (3)

Criterion	Justification

Course Outcome: "CO1" / Cognitive Level: "Understand"

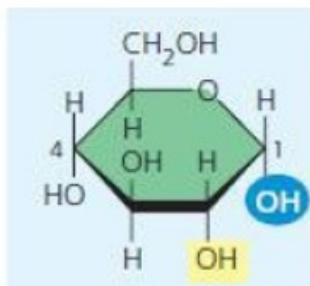
- 1B) Apply the principles of protein denaturation to explain which of the four structural levels of proteins-primary, secondary, tertiary, or quaternary-is most susceptible to denaturation and why (Provide most susceptible two levels). (3)
Which level is less affected? Justify

Protein Structure Level	Stabilizing Forces	Susceptibility to Denaturation	Explanation

Less affected structure :

Course Outcome: "CO1" / Cognitive Level: "Application"

- 1C) Here is a monomer. (4)



- (i) Illustrate the structure of the polymer formed from the above monomer using a figure.
(ii) Analyze the structural and functional features of the resulting polymer.
(2+2 = 4 marks)

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2A) Draw and interpret a Punnett square for a dihybrid test cross involving traits for flower color (Pp) and plant height (Tt) crossed with a plant of genotype pptt. What are the possible genotypes of the offspring? Dominant characters: Purple and Tall. Recessive: White and Dwarf (3)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- 2B) i) Calculate the recombination frequency between factors for body colour and wing shape. (Gray body colour and normal wing are dominant) in a hypothetical organism, given the following offspring from a test cross: (3)

- 965 gray-normal
- 944 black-vestigial
- 206 gray-vestigial
- 105 black-normal

- 185 black-normal

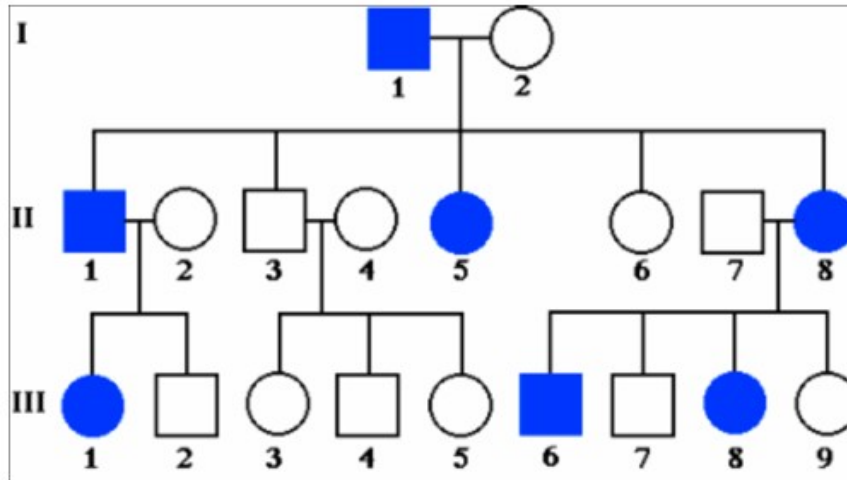
ii) What does this frequency suggest about the linkage between these factors?

Course Outcome: "CO2" / Cognitive Level: "Application"

2C)

Evaluate the pedigree chart and answer the questions

(4)



- Most probable mode of inheritance
 - Justify your reasoning
- Find the genotype of individual 6 and 8 in third generation
(1+2+1 = 4 marks)

Course Outcome: "CO2" / Cognitive Level: "Evaluate"

3A)

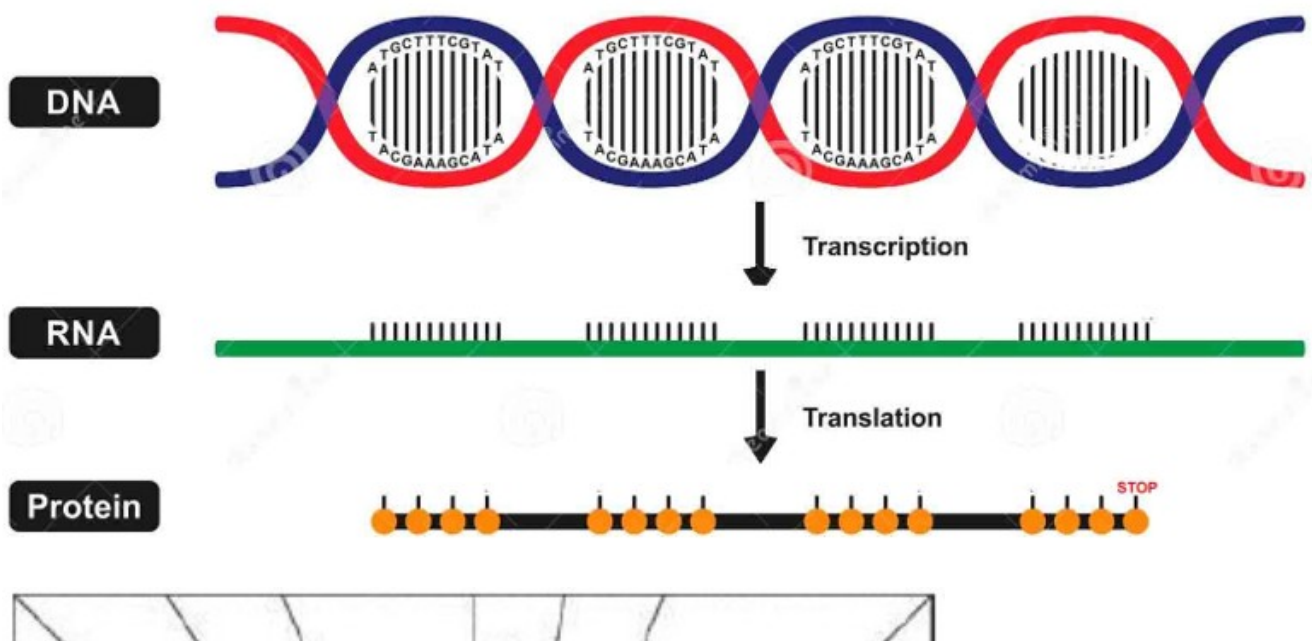
Calculate the proportion of heavy, hybrid, and light DNA strands after three generation of replication in an alien bacteria originally grown in ^{15}N and then transferred to ^{14}N medium. Show your calculation. Mode of replication = Semi conservative

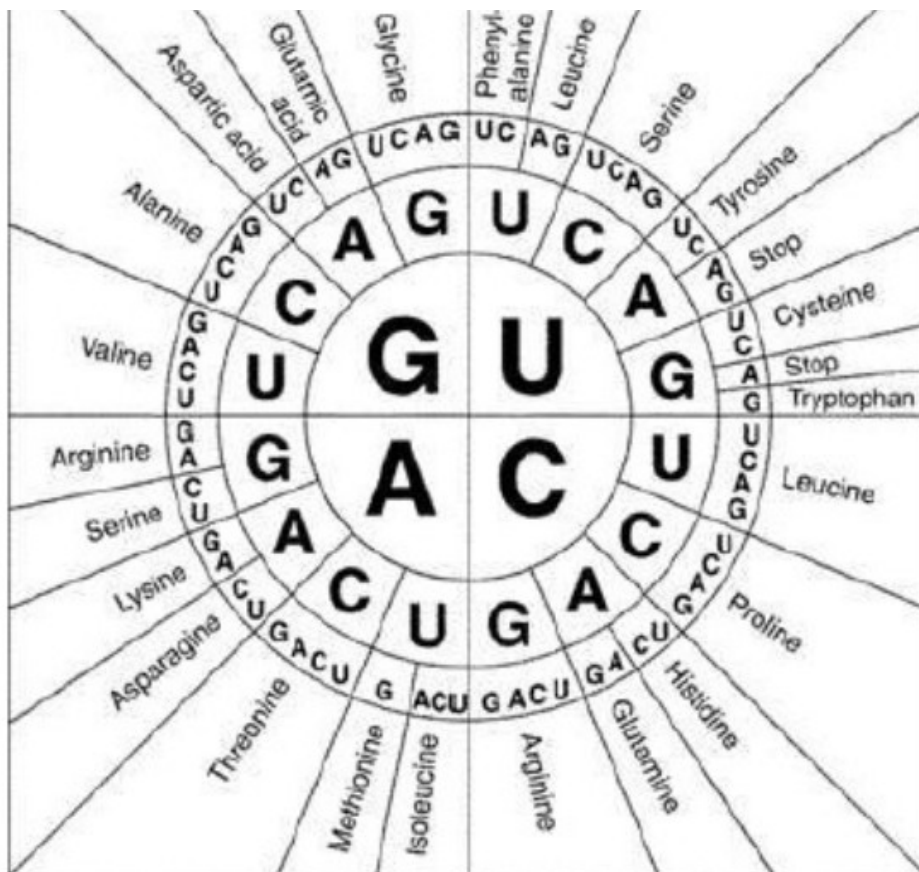
(3)

Course Outcome: "CO3" / Cognitive Level: "Application"

3B)

(4)





Evaluate the above figure and answer the following questions.
The DNA produces an mRNA with a proper start and stop

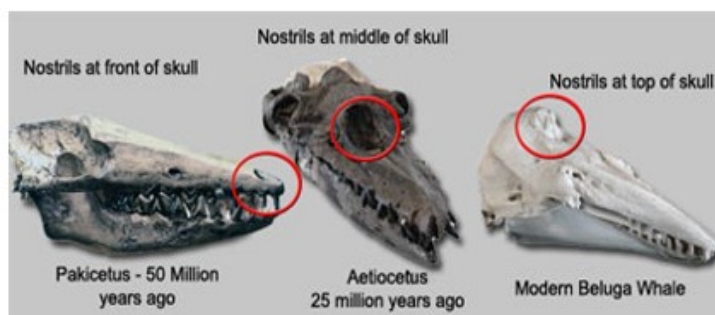
- Label the DNA ends
- Construct the mature RNA
- Label the sequence in DNA corresponding to stop codon of mRNA.
- Construct tRNA anticodon for the second amino acid and label the ends

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 3C) Differentiate between the roles of ^{35}S and ^{32}P in the experiment. If Hershey and Chase were baking a DNA cake, which ingredient would they label to prove it's not just frosting (protein)? Analyze (3)

Course Outcome: "CO3" / Cognitive Level: "Evaluate"

- 4A) Explain the concepts; (3)
- Charles Darwin and the Madagascar Orchid (1.5M)
 - +/+; +/- and -/- interactions (1.5M)
- 4B) How does the nasal drift of nostrils in vertebrates illustrate evolutionary adaptation? (3)



Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 4C) "Mutations acting as a system to improve existing designs" Analyze by providing examples (4)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5A) Nature has evolved efficient solutions - humans can mimic these to improve design, engineering, and innovation. Analyze the statement by providing logical examples discussed during your class. (3)

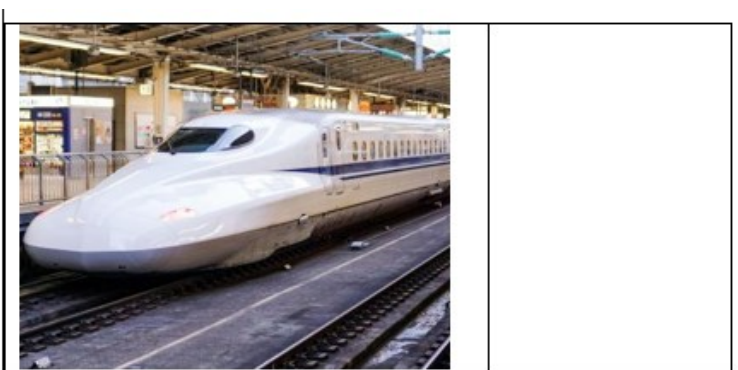
statement by providing logical example discussed during your class

- 5B) The adaptive features of the Australian Devil Lizard helps to survive in desert conditions. How can this be helpful in biomimicry? (3)



Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5C) Give one example of living organisms which Bioinspired to create the following technology. (4)



Course Outcome: "CO4" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 BIO_1071:BIOLOGY FOR ENGINEERS

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	14
Total			44

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 BIO_1071:BIOLOGY FOR ENGINEERS

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	2	6(13.64%)
3	Application	3	9(20.45%)
4	Analysis	5	18(40.91%)
5	Create	3	11(25%)
6	Evaluate	0	0(0%)
Total		13	44 (100%)